

EXAM 7 – SPRING 2011

18. (2.5 points)

An insurer that writes only commercial property insurance wants to model its underwriting risk. The carrier currently writes in only five states.

Over the next five years the carrier plans to:

- Begin writing commercial auto, excess umbrella and workers compensation insurance.
- Expand into more states.
- Sell more insurance directly through the internet, thereby eliminating some broker commissions.
- Significantly increase loss control efforts to mitigate losses with a corresponding increase in expenses.

a. (1 point)

For each of the four planned expansions, briefly discuss how the firm can model its underwriting risk using best practices.

b. (0.5 point)

Describe a risk correlation that should be reflected in the modeling of underwriting risk for this firm.

c. (0.5 point)

Describe the impact of not reflecting the correlation identified in part b. above in the model.

d. (0.5 point)

Describe a latent risk that should be reflected in the modeling of underwriting risk for this firm.

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Sample Solutions
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Solution 1

a)

For first two ventures, candidate could talk about pricing, claim issues and that model should account for:

- 1) Attritional claims – the risk that there is a higher frequency of small claims or that these claims have a higher average cost than expected.
- 2) Large claims – the risk that there are more large claims than expected or that the large claims received are more severe than expected.
- 3) Catastrophes – the risk that a catastrophe may occur causing higher claim frequency and/or severity than expected.
- 4) correlations, latent exposures.

For each of third and fourth expansions, candidate should recommend that model accounts for an expense or pricing issue such as:

- Is product priced to reflect changing expenses and/or market pressures.
 - Proportionate expenses such as commission should be modeled directly.
 - Profit commissions should be simulation specific.
 - Over time expense levels will be linked to inflation and the model should reflect this.
- There may be other acceptable answers.

b)

any of the following possible answers:

- There may be correlations between lines
- Attritional losses, particularly from longer tail business. These claims will often involve bodily injury and are sometimes subject to legislative change, whereby all cases settled after a particular date will be materially higher than previously experienced. This risk will correlate across years and business classes, affecting both underwriting and reserving risk. It is important that such “shocks” are modeled and that appropriate correlations are captured.
- Cross-class correlations that need to be captured. In particular a number of classes, otherwise uncorrelated, may exhibit some dependency in the tail of their distributions. Such dependency may not be contained in the available data so estimates of the level of the dependency will need to be chosen using considerable

c)

Answer should follow from what they selected in b and discuss result of not reflecting correlation of either double-counting an impact or underestimating one.

d)

Examples such as asbestos, other mass tort excess umbrella claims, comp occupational hazards etc., are acceptable. Candidate should also describe issues with estimating severity and likelihood of such an event happening.